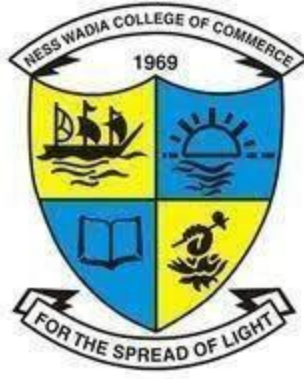




DEPARTMENT OF  
**BACHELOR BUSINESS ADMINISTRATION (COMPUTER APPLICATION)**

NESS WADIA COLLEGE OF COMMERCE, PUNE

ACADEMIC YEAR 2025-256

SEMESTER-VI PROJECT REPORT ON

**QR CODE ORDERING SYSTEM**

SUBMITTED TO

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**UNDER THE GUIDANCE OF**

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Prof. ASHWINI WHAGMARE

# Acknowledgment

I sincerely acknowledge the development of the **QR Code Based Food Ordering System**, which aims to simplify and modernize the food ordering process by connecting customers, shop owners, and administrators efficiently. This system provides a centralized platform where customers can scan QR codes, view menus, and place orders directly, while shop owners can manage menus and orders in an organized manner.

The successful completion of this project reflects a combination of effective system design, smooth functionality, and user-friendly features tailored to meet the needs of both customers and shop owners. The system ensures a seamless ordering experience and improves operational efficiency.

The integration of key modules such as QR code generation, digital menu management, order processing, and real-time updates ensures a structured and efficient workflow within the system.

This project represents a step towards digitizing and improving traditional food ordering systems, making them more efficient, contactless, and accessible for all users.

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# 1. INTRODUCTION

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## 1.1 Problem Definition

In traditional restaurants and food outlets, the ordering process is mostly handled manually, where customers depend on waiters to place their orders. Although this method has been used for a long time, it suffers from several inefficiencies and limitations, especially during busy hours.

The manual ordering system creates problems such as delays, incorrect orders, lack of coordination, and poor customer experience. With the increasing demand for fast and digital services, there is a need for a more efficient and automated solution.

## Major Problems Identified

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### 1. Inefficient Communication

In a traditional system, communication between customers, waiters, and kitchen staff is not properly structured.

- Waiters may misunderstand customer orders
- Verbal communication can lead to confusion
- Orders may not reach the kitchen accurately
- Customers are not informed about order status

**Result:** Miscommunication leads to wrong orders and dissatisfaction.

### 2. Manual and Time-Consuming Processes

The entire ordering process depends on manual work.

- Waiters take orders using pen and paper
- Orders are manually transferred to the kitchen
- Billing is done separately
- High dependency on human effort

**Result:**

- Increased waiting time
- Slower service during peak hours
- Inefficiency in handling multiple orders

### **3. Lack of Centralized Order Management**

There is no proper system to store and manage orders.

- Orders are not recorded digitally
- Difficult to track previous orders
- No proper system to manage multiple tables

**Result:**

- Loss of data
- Poor order management
- Difficulty in analyzing sales

### **4. Limited Access to Menu Information**

Customers depend on physical menus.

- Menu may not be updated regularly
- Limited information about items
- No images or descriptions available

**Result:**

- Poor user experience
- Customers may not explore full menu

### **5. Human Errors**

Manual systems are prone to errors.

- Wrong items may be ordered

- Missing items in order
- Incorrect billing

**Result:**

- Customer dissatisfaction
- Financial loss for business

## **1.2 Need for the New System**

To overcome the limitations of the traditional system, a **QR Code Based Food Ordering System** is required.

This system allows customers to scan a QR code using their smartphones and directly place orders without the need for waiters.

### **Reasons for Developing the System**

#### **1. Automation of Ordering Process**

- Customers can place orders themselves
- Reduces dependency on waiters
- Faster service

#### **2. Centralized Order Management**

- All orders are stored in a database
- Easy tracking and management
- Helps in generating reports

#### **3. Improved Communication**

- Orders directly go to the system

- Eliminates miscommunication
- Accurate order processing

#### **4. Contactless and Hygienic System**

- Customers scan QR codes instead of using menus
- Reduces physical contact
- Safer and hygienic

#### **5. Better Customer Experience**

- Quick ordering
- Easy menu navigation
- Reduced waiting time

#### **6. Business Efficiency**

- Handles multiple orders easily
- Improves workflow
- Reduces manpower cost

### **1.3 Project Scope**

The QR Code Food Ordering System is designed to manage food ordering digitally across multiple shops.

#### **1. Customer Module**

- Scan QR code assigned to a shop
- View digital menu
- Add items to cart
- Place order

#### **2. Shop Owner Module**

- Add, update, and delete menu items

- View incoming orders
- Manage order status

### **3. Admin Module**

- Register and manage shop owners
- Generate unique QR codes for each shop
- Monitor overall system activity

### **4. System Features**

- Unique QR code per shop
- Real-time order processing
- Secure login system
- Multi-shop support
- Database management



## **2. ANALYSIS**

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### **2.1 Feasibility Study**

#### **1. Technical Feasibility**

The system is technically feasible because it uses commonly available technologies and tools.

#### **Technology Stack**



- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP
- **Database:** MySQL
- **QR Code Generation:** Library/API

## **System Architecture**

- Client-Server Architecture
- Customers access system via mobile browser
- Server processes requests and stores data

## **Technical Features**

- QR code generation for each shop
- Secure login and authentication
- Real-time data processing
- Scalable system

## **Advantages**

- Easy to develop and maintain
- Works on all smartphones
- No need for additional devices

## **2. Economic Feasibility**

The system is economically feasible and budget-friendly.

## **Cost Factors**

- Development cost
- Hosting cost
- Maintenance cost

## **Benefits**

- Reduces labor cost
- Improves efficiency
- Increases customer satisfaction
- Suitable for small businesses

## **Return on Investment**

- Faster service leads to more customers
- Increased sales
- Reduced operational costs

## **3. Operational Feasibility**

The system is easy to use and can be implemented smoothly.

## **User Acceptance**

- Customers: Easy to use (scan and order)
- Shop owners: Simple dashboard
- Admin: Full control system

## **Training Requirement**

- Minimal training required
- Simple user interface
- Easy navigation

## **System Operation**

- Works through web browser
- Accessible anytime and anywhere
- No complex setup

## 2.2 Hardware and Software Requirements

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### Hardware Requirements

- Processor: Intel i5 or above
- RAM: Minimum 8GB
- Storage: 500GB
- Smartphone for customers
- Internet connection

### Software Requirements

- Operating System: Windows / Linux
- Frontend: HTML, CSS, JavaScript
- Backend: PHP
- Database: MySQL
- Browser: Google Chrome / Microsoft Edge

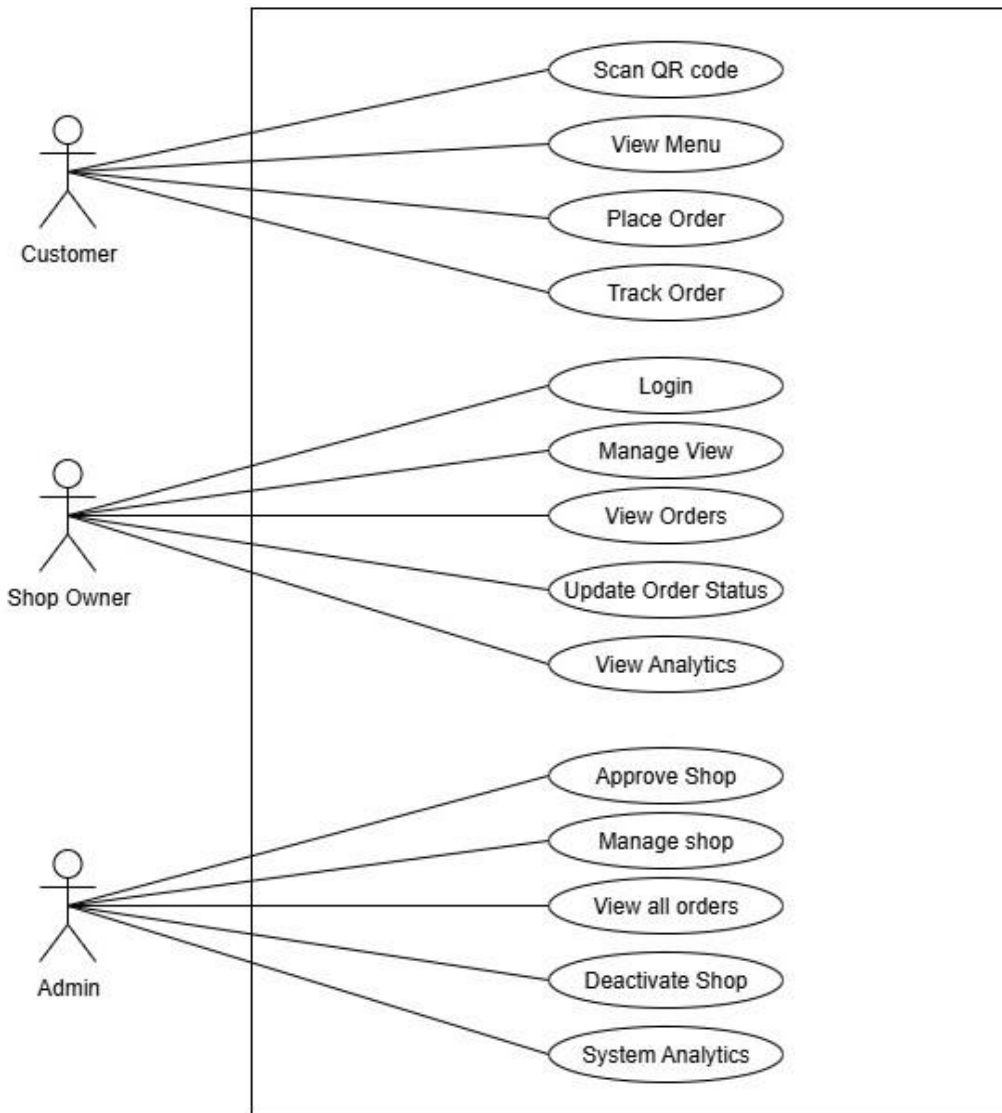
## 3. DESIGN

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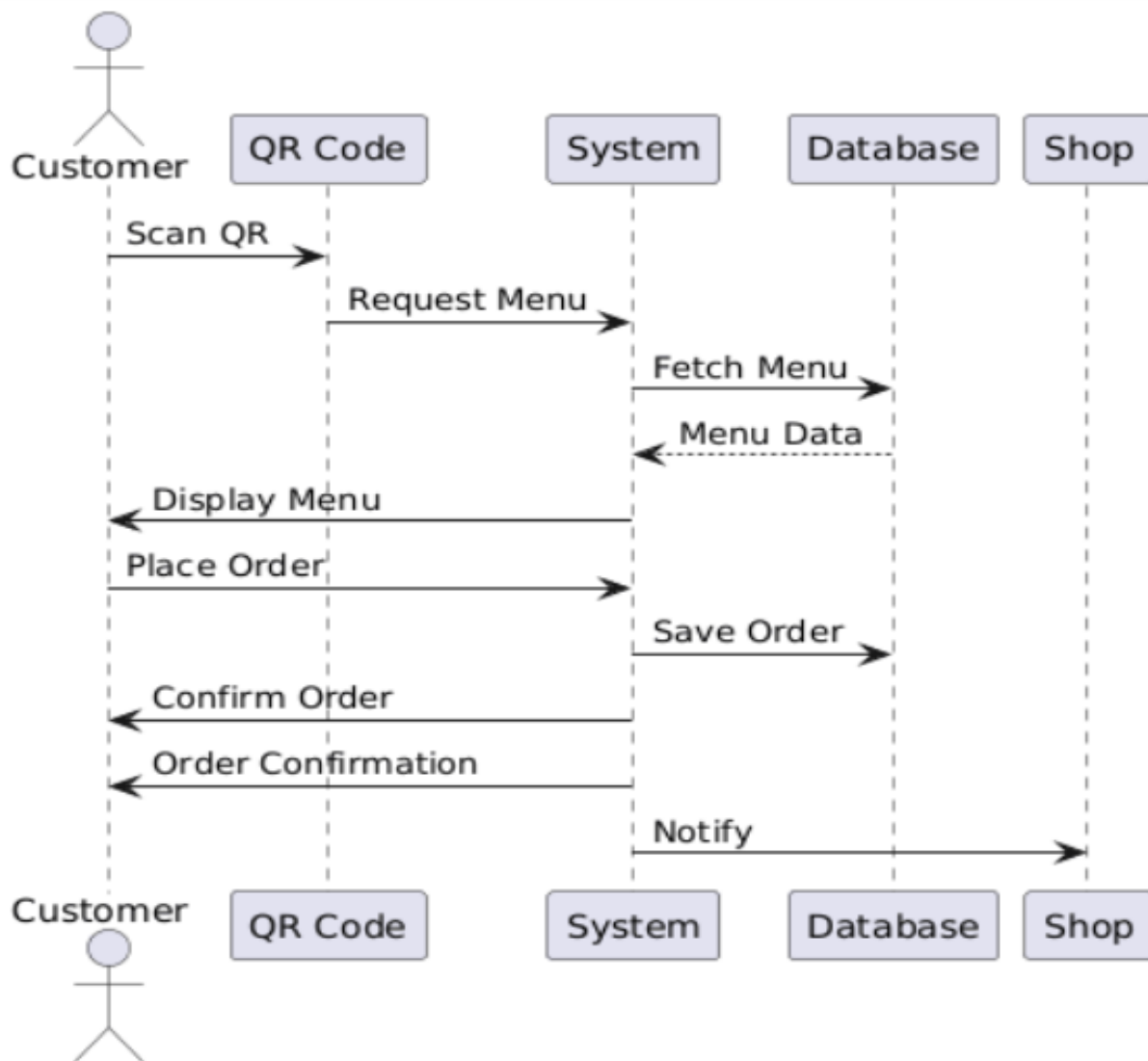
### 3.1 UML DIAGRAMS

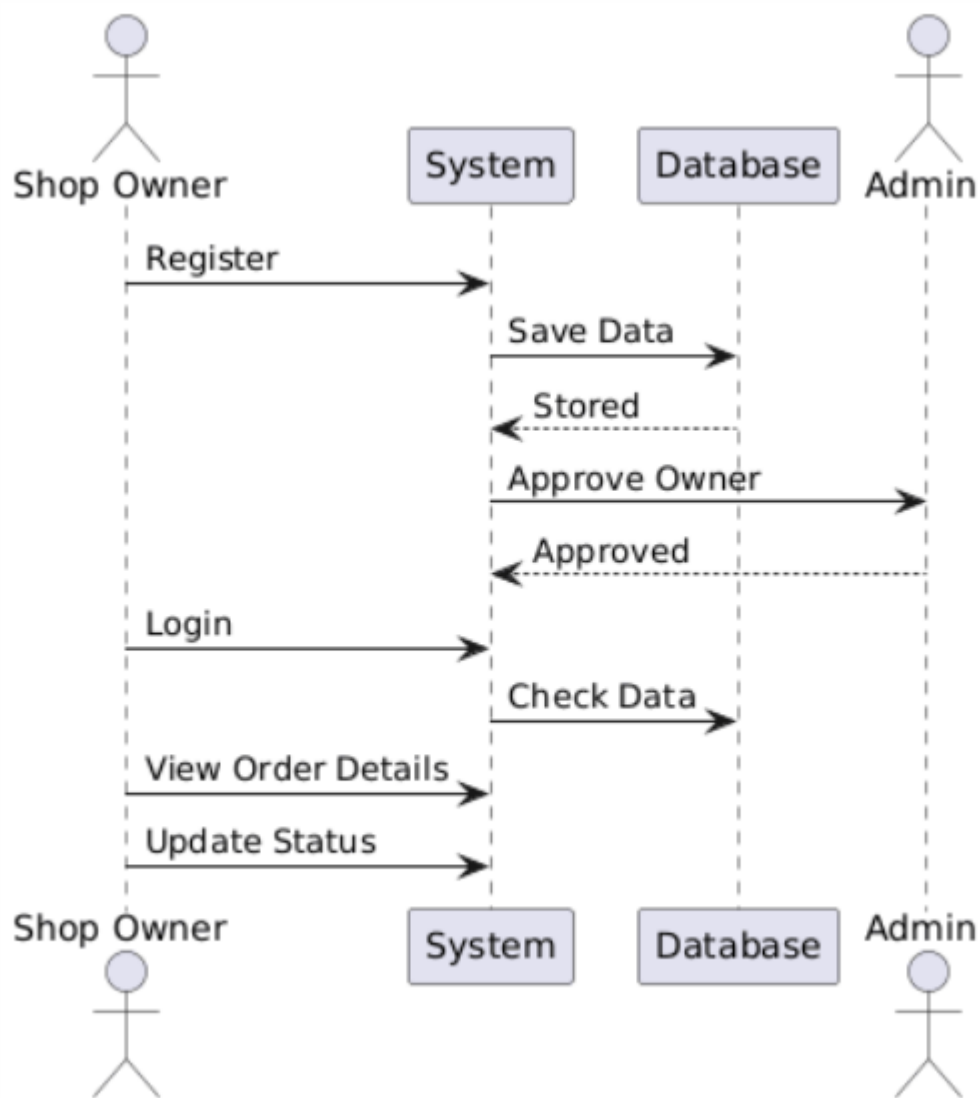
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#### 1. Use Case Diagram

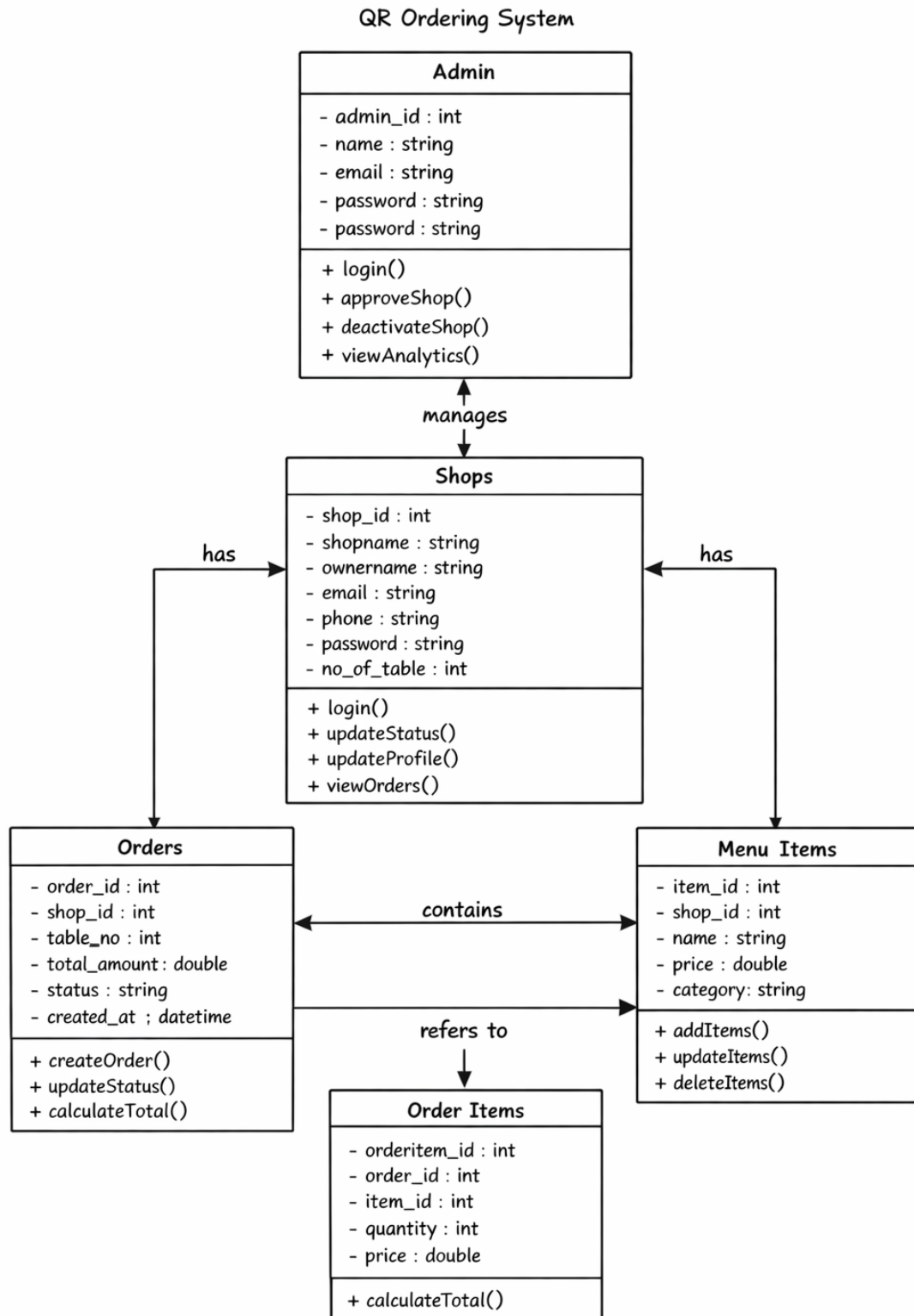


## 2. Sequence Diagram



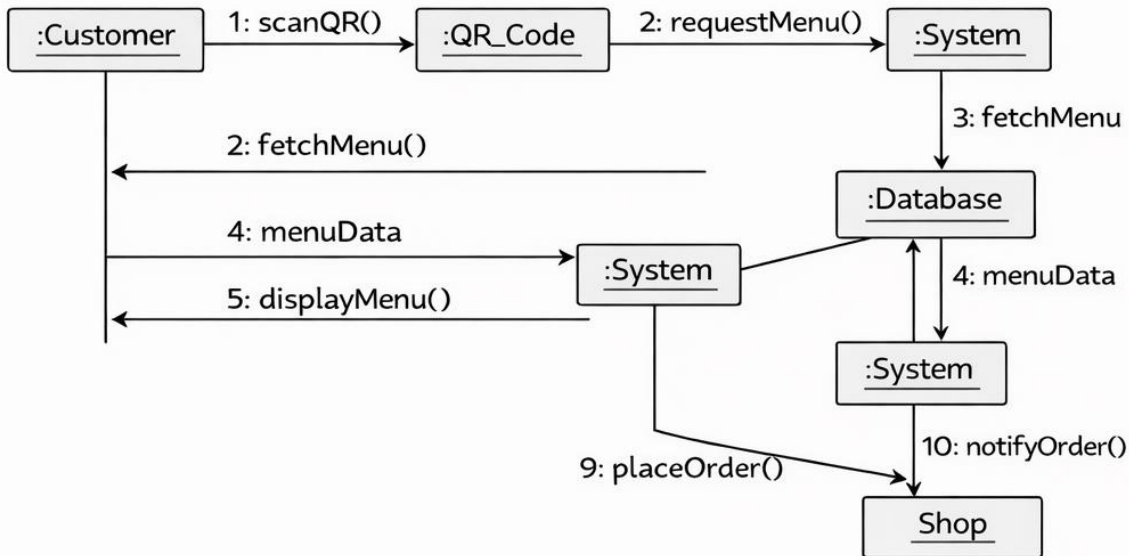


### 3. Class Diagram

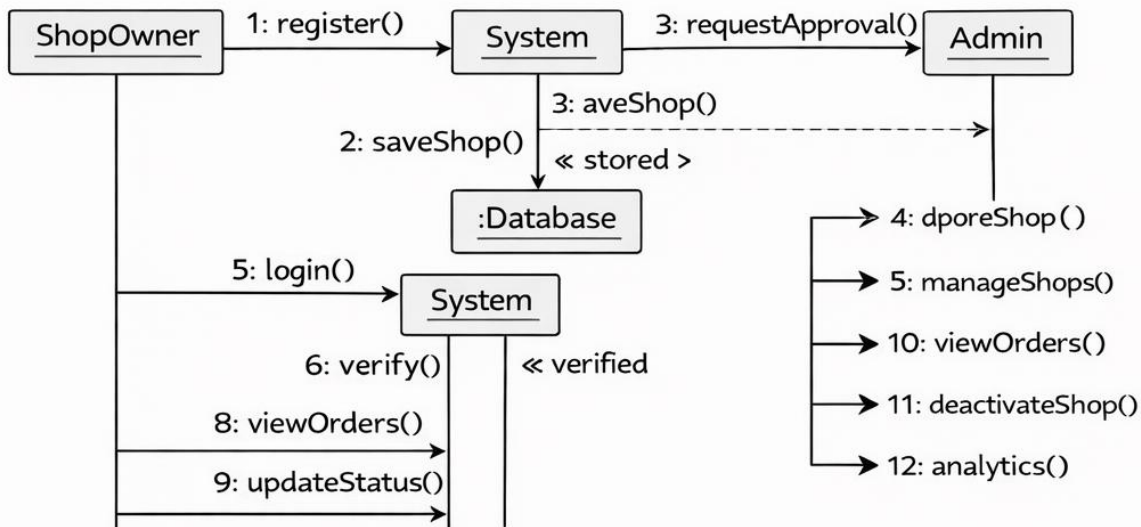


## 4. Deployment Diagram

### 1. CUSTOMER QR ORDERING – COLLABORATION DIAGRAM

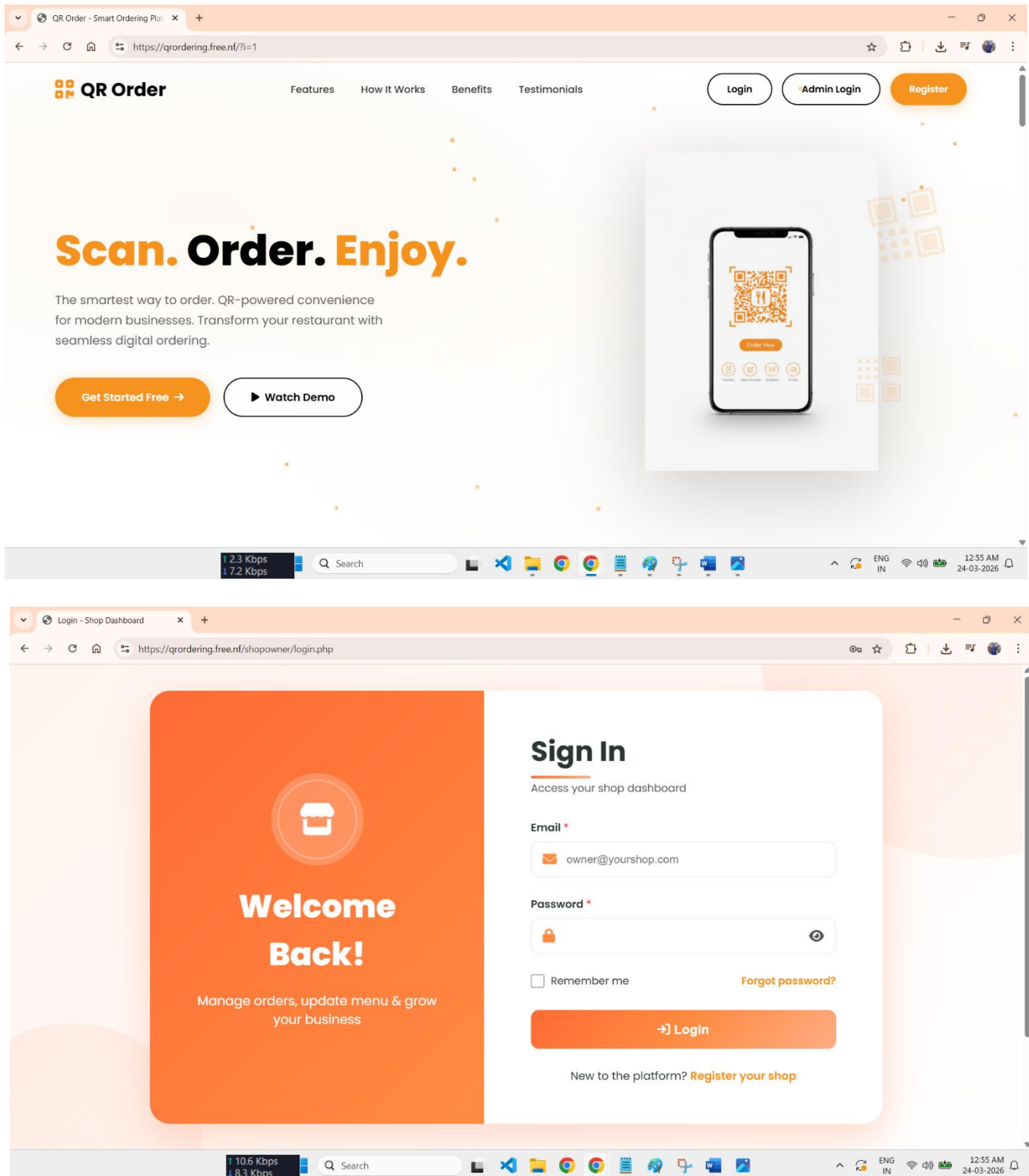


### 2. SHOP OWNER + ADMIN – COLLABORATION DIAGRAM





## 4. Screenshot



Live Orders - puri

https://qrordering.free.nf/shopowner/dashboard.php

 puri

Dashboard

Menu

Orders

Analytics

Settings

P

puri

Owner

Profile

Logout

Live Orders

Manage your restaurant in real-time

7 PENDING

1 COOKING

7 READY

Active Orders

#19

11:25 PM

SERVED

# 09

Burger1₹150

Cheese Pizza1₹180

pepsi1₹20

Total₹350

Ready

#18

11:25 PM

PENDING

# cd4df803b8ee02db89d90d8fa915b427

Burger1₹150

Cheese Pizza1₹180

pepsi1₹20

Total₹350

Update Status

#17

07:57 PM

SERVED

# 9812

Burger1₹150

Total₹150

Ready

Refresh

0.9 Kbps

0.9 Kbps

Search



ENG

IN

12:55 AM

24-03-2026

Menu Management - puri

https://qrordering.free.nf/shopowner/menu.php

 puri

Dashboard

Menu

Orders

Analytics

Settings

P

puri

Owner

Profile

Logout

Menu Management

Add, edit or remove menu items

5 TOTAL ITEMS

₹112 AVERAGE PRICE

+ Add New Item

ITEM NAME \*

e.g., Chicken Burger

PRICE (₹) \*

0.00

+ Add Item

Clear

All Items (5)

3.9 Kbps

17.5 Kbps

Search

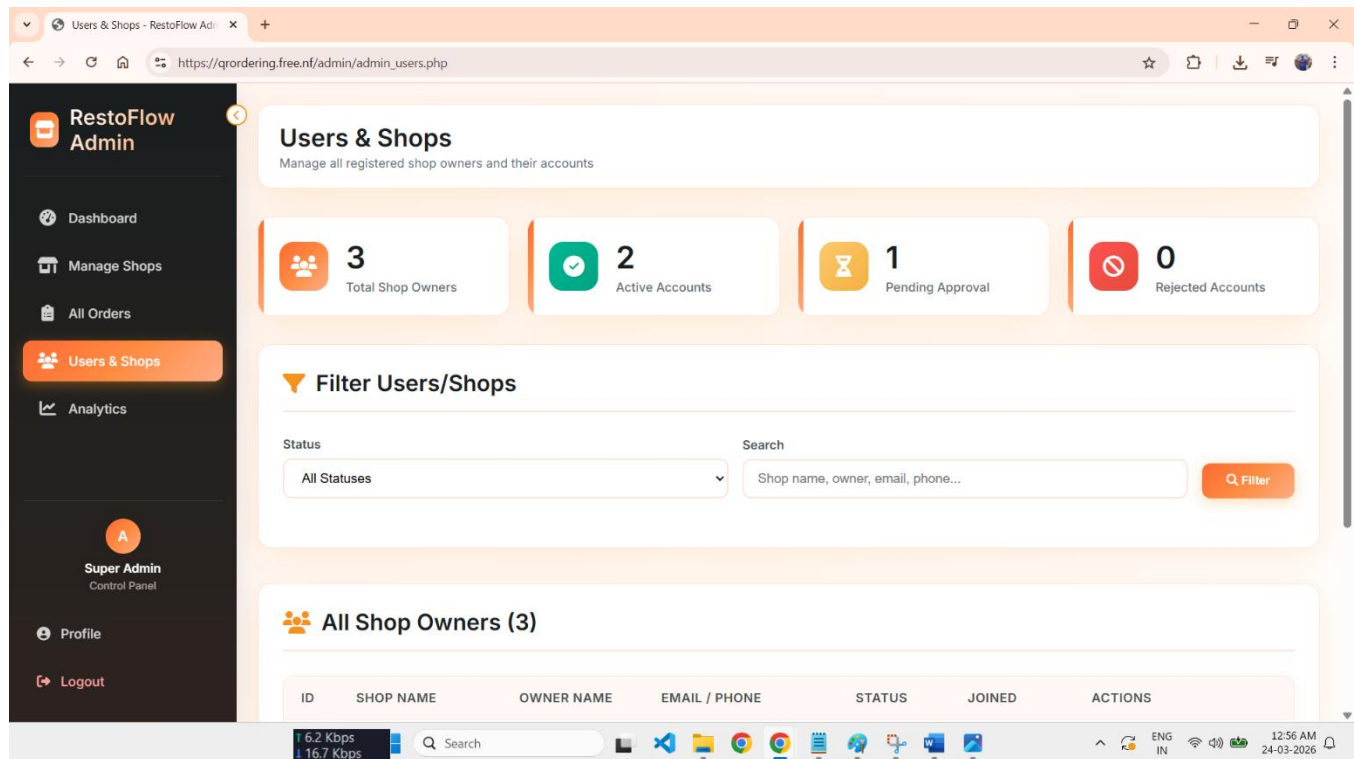
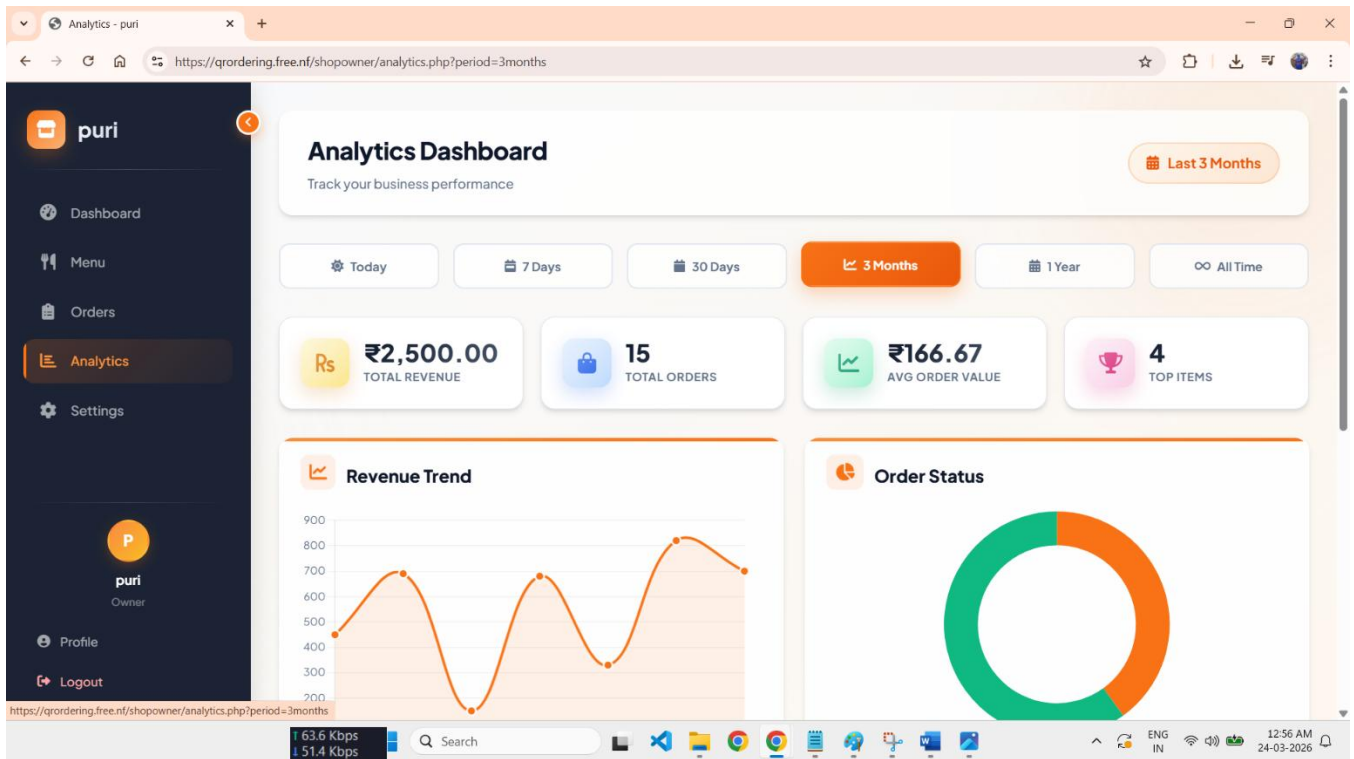


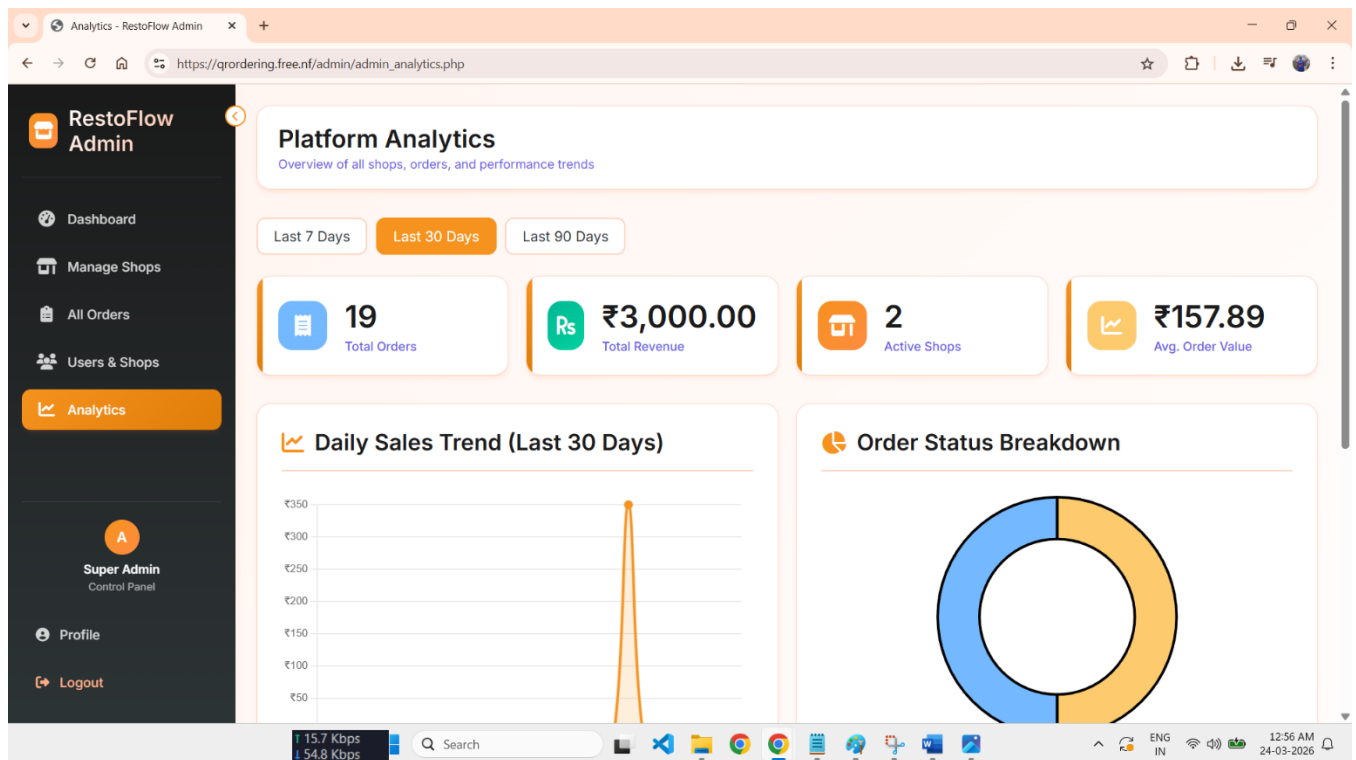
ENG

IN

12:55 AM

24-03-2026





Profile - puri

https://qrcodeing.free.nf/shopowner/profile.php

### puri

Dashboard Menu Orders Analytics Profile

15 TOTAL ORDERS

₹3,820 REVENUE

5 MENU ITEMS

4.8 RATING

#### Edit Profile

Shop Name: puri

Owner Name: mahesh

Email Address: mahesh@gmail.com

Phone Number: 8698636924

Account Status: Active

Member Since: February 09, 2026

Save Changes

#### QR Code

Customers can scan this code to view your menu and place orders directly.

Download QR Code

#### Change Password

2.0 Kbps 1.5 Kbps

Search

ENG IN 12:57 AM 24-03-2026

## 5. LIMITATIONS

Although the QR Code Food Ordering System improves efficiency and customer experience, it has certain limitations:

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### 1. Dependency on Internet Connection

#### Explanation:

The system is completely web-based and requires an active internet connection for both customers and shop owners.

#### Impact:

- Customers cannot place orders without internet
- Orders may fail during network interruptions
- Poor connectivity can delay order processing

### 2. No Offline Functionality

#### Explanation:

The system does not support offline mode.

#### Impact:

- Cannot be used in areas with weak or no network
- Limits usability in remote locations

### 3. Limited Payment Integration

#### Explanation:

The system may only support basic or manual payment methods (like cash).

#### Impact:

- No support for UPI, cards, or wallets

- Reduces convenience for customers
- Limits digital transaction options

## **4. Basic Security Features**

### **Explanation:**

The system may use simple login authentication without advanced security measures.

### **Impact:**

- Risk of unauthorized access
- Data security can be improved
- No encryption or advanced authentication

## **5. No Mobile Application**

### **Explanation:**

The system is only available through web browsers.

### **Impact:**

- Less user-friendly compared to mobile apps
- Reduced accessibility
- No push notifications

## **6. Limited Customization for Shops**

### **Explanation:**

Shop owners may have limited options to customize their menu and interface.

### **Impact:**

- Less flexibility
- Same UI for all shops

## **7. No Real-Time Order Tracking**

## **Explanation:**

Customers may not be able to track their order status in real time.

## **Impact:**

- Customers do not know order progress
  - Reduced transparency
- 

# **6. FUTURE ENHANCEMENTS**

The system can be further improved by implementing the following features:

## **1. Online Payment Integration**

### **Description:**

Integrate digital payment systems.

### **Features:**

- UPI payments
- Debit/Credit card support
- Wallet integration

### **Benefits:**

- Faster transactions
- Improved customer experience
- Cashless ordering

## **2. Mobile Application Development**

### **Description:**

Develop Android and iOS apps.

**Features:**

- Push notifications
- Easy ordering interface
- Faster performance

**Benefits:**

- Better usability
- Increased engagement
- Improved accessibility

**3. Real-Time Order Tracking****Description:**

Allow customers to track order status.

**Features:**

- Order received
- Preparing
- Ready to serve

**Benefits:**

- Transparency
- Improved customer satisfaction

**4. AI-Based Recommendation System****Description:**

Use AI to suggest food items.

**Features:**

- Based on previous orders
- Popular items suggestion

**Benefits:**



- Increased sales
- Personalized experience

## **5. Multi-Language Support**

### **Description:**

Support multiple languages in UI.

### **Features:**

- Language selection option
- Regional language support

### **Benefits:**

- Better accessibility
- User-friendly for all customers

## **6. Advanced Analytics for Shop Owners**

### **Description:**

Provide data insights.

### **Features:**

- Sales reports
- Popular items analysis
- Customer trends

### **Benefits:**

- Better business decisions
- Increased profitability

## **7. Notification System**

**Description:**

Add notification system.

**Features:**

- Order confirmation
- Order status updates
- Alerts for shop owners

**Benefits:**

- Real-time communication
- Improved efficiency

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## 7. REFERENCES AND BIBLIOGRAPHY

The development of the QR Code Food Ordering System is supported by the following resources:

### 1. PHP Documentation

**Description:**

Official PHP documentation for backend development.

**Usage:**

- Database connectivity
- Server-side scripting
- Form handling

### 2. MySQL Documentation

**Description:**

Provides guidance on database management.

**Usage:**

- Table creation
- Data storage
- Query execution

### **3. QR Code Libraries**

**Description:**

Libraries used for generating QR codes.

**Usage:**

- Generating unique QR codes for each shop
- Linking QR codes to menu pages

### **4. W3Schools**

**Description:**

Online learning platform for web development.

**Usage:**

- HTML, CSS, JavaScript concepts
- Form validation
- UI design

### **5. GeeksforGeeks**

**Description:**

Programming tutorials and examples.

**Usage:**

- Logic building
- Backend concepts
- Database queries

## **6. Additional References**

- Web development tutorials
- Online forums and documentation
- Practical implementation experience

